

- You are allowed to increase the number of CPUs in each node from one to four.
- You are allowed to increase the physical memory of each node from 8GB to 16GB.
- You are allowed to use a different network communication channel, which has lower network latency.
- You are allowed to use a different network communication channel, which has a higher network bandwidth.
- You are allowed to replace the HDDs in each node with SSDs.

If you can choose only one of the above options, which would you choose? Explain the rationale behind your choice. Is your choice application dependent?

- Can you explain the difference between linear regression and logistic regression? Can you use a linear regression technique to solve a logistic regression problem?
- Let us consider the problem of predicting house prices in Bengaluru. Consider that your data set has a large number of features, say around 10,000. But you have training data only for 100 data points.
 - How would you address the problem of overfitting the model to the limited training data?
 - Are you familiar with L1 and L2 regularisation? What is the difference between the two?
- What are the different machine learning algorithms that you have used? Can you compare decision trees, neural networks and support vector machines?
- What is the well-known 'kernel trick' in machine learning? (We are not talking about the Linux kernel here.)
- You are given a set of newspaper articles, which you have to categorise into four different categories such as politics, entertainment, sports and miscellaneous. What classification algorithm would you use to solve this problem for the following two cases?
 - Your data set contains training data, for which you are given labelled categories for a set of documents.
 - Your data set does not have any labelled instances.
- Let us consider a binary classification problem in which you are asked to label emails as spam or non-spam. You are given a large enough training data set (100,000 emails), where the emails are already classified as spam or non-spam. Using this training data, you are asked to train a binary classifier, which can mark an incoming email as spam or non-spam. However, there is one problem with the training data that you have been given. Among the 100,000 emails, only 100 are labelled as spam. All other training data are of the non-spam category. What would be the problem if you train your model with such an unbalanced training data set? How would you address this issue?
- We have all heard a lot about IBM's Watson system beating human contestants in the TV question and answer game 'Jeopardy'. Watson employs what is known as a 'deep

parsing' pipeline for text processing. What is the difference between shallow parsing and deep parsing in natural language processing?

- Consider the following two sentences:

- John bought a car.
- A car was bought by John.

Can a shallow parsing system figure out that the two sentences are semantically equivalent? If not, what part of the NLP can determine that these two sentences are equivalent?

- You are all familiar with product review systems used in various websites such as Amazon product reviews, TripAdvisor's review of hotels or Yelp's review of restaurants. We also know that many of the reviews can be fake, having been seeded by sellers themselves or by a competitor (if it is an adverse review). How would you design a system to filter only the 'useful' reviews for a given product? While filtering out fake reviews is a basic necessity, among the non-fake reviews, how would you give different weightage to different users?
 - You are given a large text document and asked to create a summary of it. There are two types of summarisation algorithms, namely, extractive vs abstractive summarisation. In extractive summarisation, the summary would contain sentences extracted from the text. In abstractive summarisation, the summary would contain the gist of the original text, without including the exact sentences from the text. What kind of summarisation algorithm would you choose? Explain the rationale behind your choice. Now assume that there is a real-time conversation that is going on. How would you create a running summary of it? If you are interested in doing some data science experiments, Kaggle (www.kaggle.com) is a good place to start. There are generally a number of active competitions on, some of which are for learning purpose only. Currently, it's running a competition on sentiment analysis of movie reviews, which you may want to enter (<https://www.kaggle.com/c/word2vec-nlp-tutorial>).
- If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Till we meet again next month, happy programming! 

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